EXERCISE 11.5

(1) (i) $\frac{160}{729}$ (ii) $210\left(\frac{1}{5}\right)^4\left(\frac{4}{5}\right)^6$ (iii) $\binom{9}{7}\left(\frac{1}{2}\right)^7\left(\frac{1}{2}\right)^2$ (2) (i) $\binom{10}{4}\left(\frac{1}{4}\right)^4\left(\frac{3}{4}\right)^6$ (ii) $1-\frac{3^{10}}{4^{10}}$

(3) (i) 50, 25 (ii) 40, $\frac{100}{3}$ (4) $\frac{270}{1024}$ (5) (i) $1-0.95^{10}$ (ii) $\binom{10}{2}(0.05)^2(0.95)^8$

(6) (i) $\binom{12}{10}(0.9)^{10}(0.1)^2$ (ii) $2.1(0.9)^{11}$ (iii) $1-\left[2.1(0.9)^{11}\right]$

(7) (i) $\binom{18}{x}\left(\frac{1}{3}\right)^x\left(\frac{2}{3}\right)^{18-x}$ (ii) $\binom{18}{3}\left(\frac{1}{3}\right)^3\left(\frac{2}{3}\right)^{15}$ (iii) $1-\frac{20}{3}\left(\frac{2}{3}\right)^{17}$

(8) $\binom{6}{x}\left(\frac{1}{3}\right)^x\left(\frac{2}{3}\right)^{6-x}$, 2, $\frac{2}{\sqrt{3}}$ (9) 1, $\frac{4}{5}$

EXERCISE 11.6

1	2	3	4	5	6	7	8	9	10
(2)	(4)	(2)	(4)	(4)	(2)	(4)	(3)	(2)	(1)
11	12	13	14	15	16	17	18	19	20
(4)	(4)	(1)	(2)	(1)	(1)	(4)	(4)	(2)	(1)

Chapter 12

Exercise 12.1

1. (i) Yes, $*$ is binary on $\mathbb{R}$ (ii) Yes, $*$ is binary on A

   (iii) No, $*$ is not binary on $\mathbb{R}$

2. No, $*$ is not binary on $\mathbb{Z}$ 3. $\frac{-88}{15}$

4. Yes, usual multiplication is binary on A

5. (i) The given operation $*$ is closure and commutative but not associative on $\mathbb{Q}$.

   (ii) Identity does not exist and so inverse does not exist.

6.

$*$	a	b	c
a	b	c	a
b	c	b	a
c	a	a	c

7. No. The given operation is not commutative and associative

8. (i) $A \vee B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$ (ii) $A \wedge B = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$

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